Criterion	Unit · source	S275N	Al 2618-T6	Ti-8Al-1Mo-1V
Max. service temperature > 300 °C	°C · p.9	● 471–501	✖ **250–300**	● 407–455
Castable or 3-D printable	stage 2 · p.5	● yes	● yes	● yes
Fracture toughness ≥ 10	MPa·m½ · p.9	● 26–38	● 31–34	● 65–75
Allowable stress at 300 °C	MPa · σy/1.5	107	✖ **no data**	345
Young's modulus at 23 °C	GPa · p.8	200–221	72–76	121–127
Density	kg/m³ · p.8	7800–7900	2720–2780	4350–4400
Price	USD/kg · p.8	1.0–1.6	3.1–3.5	23–26
Production CO_2 (virgin + powder)	kg/kg · p.11–12	9	17	42
Life-cycle CO_2 per kg of part	kg/kg · card	346	354	379
Oxidation at 500 °C	p.11	acceptable	✖ **unacceptable**	acceptable
Stress-corrosion cracking	p.11	slight	✖ **high**	slight
Sourcing risk	p.10	low	✖ **very high**	medium
Verdict		**Reference**	**Eliminated**	**Selected**